FARMARE — Whitepaper

An engineered, delta-neutral basis strategy — market-neutral carry, at scale

Version 2.1 — July 2026. No token. No DAO. Non-custodial vault on Ethereum mainnet.

Abstract

FARMARE is a rules-based, automated yield strategy delivered as a self-custodial ERC-4626 vault on Ethereum mainnet. It runs a single, diversified **delta-neutral basis** book — long staked-ETH collateral, short an equal notional of ETH perpetual futures — harvesting perpetual funding and staking yield with no exposure to the direction of ETH, sized with adaptive, stress-capped leverage. Depositors hold vault shares priced at daily NAV and may exit at any time; no party, including the operator, can move deposited funds to a third address. There is no governance token and no DAO: the operator sets the rules, but parameter changes (fees, venues, referral share) are bound by a 48-hour on-chain timelock and hard-coded ceilings — keeper replacement and pause are immediate so a problem can be contained — and the depositor's right to redeem is unconditional. **The headline figures are a backtest on real market data, not a lived track record: the strategy was replayed over 1,008 days (Oct 2023 → Jul 2026), and the system has only been publishing forward since 14 July 2026. The vault is live on Ethereum mainnet with no third-party capital. This document is informational and is not an offer of any financial product.**

1. The problem

On-chain yield is abundant but hostile to passive capital. Stablecoin rates compress the moment capital arrives; concentrated liquidity positions earn rich fees but bleed value in any trending market ("gamma bleed"); incentive farms carry exploit and rug risk. Most retail and even institutional capital either sits in the lowest-yield venues or takes uncompensated directional and smart-contract risk chasing headline APYs. What is missing is not another high-APY venue — it is disciplined, automated *management* of the trade-off between fee income, impermanent loss, and tail risk.

2. Strategy

FARMARE Alpha runs a single, diversified **delta-neutral basis** book.

The position. It holds staked-ETH spot — liquid-staking and liquid-restaking tokens (stETH / weETH, dual-restaked on EigenLayer and Symbiotic) — and shorts an equal notional of ETH perpetual futures. The two legs cancel price exposure, so the book is **market-neutral**: it does not bet on which way ETH moves. What it earns is **carry** — the perpetual funding that longs pay shorts, plus the staking yield on the collateral, net of borrow cost.

Diversification. The book harvests a BTC/ETH/SOL basket across two funding venues (Binance, Hyperliquid). Per asset, each day, it takes the venue whose trailing 30-day funding was richer — a mechanical, ex-ante rule that keeps the system open to the better opportunity without hindsight. More assets and venues are added as they earn their place; idiosyncratic basis noise averages down, which is what lets the strategy run leverage safely.

Adaptive leverage. Because the carry is positive and the position is neutral, sizing it up multiplies return without adding directional risk. Leverage scales with the richness of trailing funding — up to a stress-capped **3x** when carry is rich, de-levering toward 1x when funding thins or stress appears. The added risk is liquidation and collateral risk, not market direction — quantified in §8.

This replaces FARMARE's earlier core–satellite design: the same disciplined, worst-regime engineering, concentrated into the highest-carry neutral strategy.

3. Where the yield comes from (and why it persists)

1. **Perpetual funding** — in crypto, perpetual longs pay shorts the large majority of the time. Measured on real data, ETH funding averaged ~14%/yr through-cycle and was positive on ~88% of days. A delta-neutral book harvests this directly, with no directional exposure.
2. **Staking + restaking yield** — the spot leg is staked and restaked (EigenLayer / Symbiotic), earning protocol issuance and AVS rewards *on top of* funding.
3. **Leverage on a positive-EV neutral book** — because the carry is positive and the position is market-neutral, sizing it up multiplies the return without adding directional risk, up to a stress-tested cap.
4. **Diversification premium** — spreading across many funding markets averages down idiosyncratic dislocation, which is what lets the book run its leverage safely.

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4. Evidence

The daily NAV series behind every chart is published as raw JSON at farmarecapital.com/nav.json, and the deployed contracts are source-verified on Etherscan. The research code that produces the backtest is **not public yet** — so these results are, for now, reproducible by us and inspectable by you, not independently re-runnable. That gap closes when the code is opened alongside the external audit.

Test	Data	Result
ETH funding harvest, through-cycle	Real funding, Binance 2019–2026	**+14.1%/yr** unlevered, positive on ~88% of days, worst rolling 12m still +1%
Real backtest of the flagship	Real BTC/ETH/SOL funding, two venues (best-of), + staking + borrow, 2023–2026	$100k → **$308k** (50%/yr, net of fees), delta-neutral through ETH's –68% drawdown
Leverage stress battery	Adversarial: depeg, venue failure, funding collapse	Safe leverage **2–3x**; 5x rejected — it liquidates on a single crisis
Diversification	Monte Carlo, 8 markets	Cuts the single-asset liquidation tail from ~13% to <1% at the same leverage
Whale on-chain validation	17 top DeBank portfolios read on-chain	Confirms staking/restaking + basis carry as the dominant real strategy

Forward expectation (through-cycle, net of fees): median ~+26%/yr, mean ~+30%. The realized 2023–2026 backtest exceeded the through-cycle median. Full risk and drawdown profile in §8.

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5. Protocol architecture

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  depositors ■■deposit/withdraw■■■  FarmVault (ERC-4626, Ethereum mainnet)
                                   · shares priced at daily NAV
                                   · venue whitelist + 48h timelock
                                   · per-venue caps · loss circuit breaker
                                   · mgmt/performance fees (HWM)
                                          ■ allocate / deallocate (bounded)
                              ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                         Venue adapters (vault ■ venue only)
                Ethena basis (sUSDe) · leveraged basis (Morpho, 3x hard cap)
                     Aave v3 / Morpho lending (safe sleeve & park)
                                          ▲
                              off-chain KEEPER executes the public algorithm
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The **keeper** is an off-chain process that signs only `allocate` / `deallocate` / `accrueFees`, exclusively between whitelisted adapters and within caps and a per-operation loss bound. It has no path to withdraw funds. Adapters can only move assets vault → venue → vault. Value (NAV) is read live from the adapters, so any loss marks down immediately and fairly.

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6. Governance — sole operator, no token

FARMARE has **no DAO and no governance token**, by explicit design. The operator (a hardware-wallet owner) is the sole decision-maker for venues, caps, fees and keeper appointment. Depositor protection is structural, not a matter of trust:

- **48-hour timelock** on fee, venue and referral-share changes — announce-then-apply, so anyone who disagrees can exit *before* it takes effect.
- **Hard-coded ceilings** the operator cannot exceed (management fee ≤5%, performance fee ≤30%).
- **Unconditional exit** — `redeem()` at NAV works at any time, even while new deposits are paused; withdrawals auto-unwind venue positions.
- **Minimal keeper** — the automation can rebalance but never withdraw.

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7. Fees

Share-based, identical to a classic fund: a management fee accrued pro-rata on AUM and a performance fee charged only on gains **above the high-water mark**, both minted to the operator as share dilution (never taken from principal). Defaults: 1%/year management, 10% performance. All fee changes pass the 48h timelock and the hard ceilings above.

No management fee while parked (design commitment, Phase 3). When the crisis guard moves the whole book into the safe lending sleeve, the strategy is not running — the vault is holding a lending position a depositor could hold themselves. In that state charging a management fee is not justified, so the commitment is to **waive it while the book is fully parked**: you pay for active management only when there is active management. The performance fee, charged only above the high-water mark, still applies — but in a weak-funding regime the NAV rarely exceeds its prior peak, so it is typically near zero. The effect is that in a prolonged lean stretch the fund's net yield stays in line with holding the lending position directly, removing the penalty for staying invested and being positioned when the carry regime returns. The deployed v0 contract accrues the management fee at all times; waiving it while parked requires the next contract revision and is gated behind the same audit as the rest of Phase 3.

7.1 Taking your yield — compound or withdraw

The vault issues a single, accumulating ERC-4626 share. Your yield lives in the share price: hold, and it compounds; redeem, and you realize it. There is no separate "income" share class in the deployed contract — one share type, one behaviour.

Because `redeem()` is permissionless and always open, a depositor who wants income rather than compounding simply withdraws a portion whenever they choose — there is no lock, no schedule, and no counterparty to ask. The site calculator offers a **Monthly income** view that illustrates, from the backtest, how much of each month's gain such a withdrawal would have realized under a high-water-mark rule (a losing month draws nothing; the prior peak is recovered first). It is an illustration of a manual action the depositor controls, not an automated payout the contract performs.

Automating this — crediting each depositor's gain to their wallet on a schedule — would require per-depositor high-water marks, a share-class registry, and a keeper allowance over depositor shares, none of which exist in the deployed contract and each of which introduces a trust assumption the current design deliberately avoids. It is therefore not offered as a live feature.

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8. Risk framework

FARMARE Alpha is a **leveraged** strategy. Its return is high because it takes real, bounded risks — priced explicitly in a production stress suite, not assumed away. **This is not a capital-preservation product.**

Return and drawdown profile (through-cycle, stress-tested, net of fees):

Median annual return	~+26%
Mean annual return	~+30%
Worst 5% of years	~–7%
Average maximum drawdown	~–11%
Severe combined crisis (funding collapse + basis dislocation + collateral depeg at once)	~–**24%** in a year

Principal risks and mitigations:

- **Funding collapse** — if perpetual funding compresses to zero or turns

negative for a sustained stretch, the levered carry bleeds. Mitigated by adaptive de-leveraging toward 1x, diversification across markets, and an automatic **crisis guard**: when trailing funding breaks down the book fully unwinds and parks in the lending sleeve until the carry regime returns — cutting the modeled combined-crisis drawdown from ~–23% to ~–10%.

- **Leverage / liquidation** — leverage is capped at a stress-tested **3x**.

5x was rejected by the stress suite: it liquidates on a single collateral depeg or venue failure. A market-wide deleveraging can still force loss.

- **Collateral (LST/LRT) depeg** — staked-ETH collateral can trade below ETH

(stETH –7% in June 2022; ezETH –10% in April 2024). Liquid-restaking exposure is capped at ≤**50%** of the collateral — the rest in stETH / plain ETH — with continuous peg monitoring and automatic unwind.

- **Venue / smart-contract** — perpetual-venue, lending-market or bridge

failure. Diversified across venues; funds sit in a non-custodial vault the operator can never withdraw from; per-venue caps bound single-venue exposure.

- **Borrow-rate spikes** — the cost of the leverage leg can spike; the model

prices spikes to 20%.

Structural protections (product-agnostic): unconditional `redeem()` at NAV, a 48h timelock on fee/venue/referral changes, hard-coded fee ceilings, per-venue caps, a per-operation and rolling cumulative-loss circuit breaker, and a fully journaled, git-committed decision history that a public dashboard rebuilds from.

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9. Security

An internal adversarial review fixed five findings — a first-deposit inflation attack, a reentrancy vector, orphaned funds on venue removal, an unbounded cumulative-loss path, and a UI decimals bug — each with a regression test. The contract suite passes 20/20 (unit, security, fuzz, and Base-mainnet fork tests against real Aave and Morpho). This internal review is **not a substitute for a professional external audit**, which is a prerequisite before third-party funds, alongside a public testnet with a bug bounty and staged deposit caps.

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10. Roadmap

- **Phase 0 — Research & backtest.** Complete. The carry is measured on real perpetual funding back to **November 2019** (~6.6 years, spanning the 2021 bull, the 2022 bear and the 2024–25 cycle); ETH spot goes back to 2016. The headline curve is a **1,008-day backtest** (Oct 2023 → today) on real funding, staking and borrow inputs: a BTC/ETH/SOL basket, each asset taking the richer of Binance/Hyperliquid by an ex-ante trailing rule. The basket policy was set on 17 Jul 2026 and the previous ETH-only curve is archived — it is a backtest, not a lived track record.

- **Phase 1 — Forward monitoring.** Active since **14 July 2026**: the same

system now runs forward on live market data, publishing every run to a public git ledger. Days of forward record so far are counted from that date, not from the start of the backtest.

- **Phase 2 — On-chain vault.** **Live on Ethereum mainnet since 15 July

2026.** FarmVault `0x873C1b701ABDb9334018f4044FEF1d10B430265C`, EthenaBasisAdapter `0x4C8B29Ed46Fc191b6C3855350653552D91FC4Fc8`, AaveV3Adapter `0xb1C01192A0eA24713C3bF36466F5a30949137B9e` — all three source-verified on Etherscan. Venues clear their 48h timelock on 17 July.

- **Phase 3 — Audit, then scale.** The Alpha-engine adapters and the

partner-referral module have **not been through an external security review**. That review, and a self-funded pilot, gate any meaningful third-party capital. The same revision introduces the parked-state management-fee waiver (§7): no management fee while the book sits fully in the safe lending sleeve.

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11. Legal & disclaimers

FARMARE does not currently custody or accept third-party funds. All performance shown is **simulated** (paper trading on live market data) and is not a forecast. On-chain deposits, once live, carry real smart-contract and DeFi risk, including total loss. Nothing herein is investment advice or an offer or solicitation of any financial product. Accepting third-party funds with management is a regulated activity (MiCA in the EU); it will not occur until the appropriate legal structuring and authorizations are in place.